**REVISION TEST****3**

Name of the Student: \_\_\_\_\_

Time: 3 hours

Class : 10<sup>th</sup>

Sub: MATHS

Sec: Orion, Nebula

Marks: 100

**I. Answer all the questions in one word or phrase.**

$12 \times 1 = 12m$

1. The prime factorisation of **1771** is

[     ]

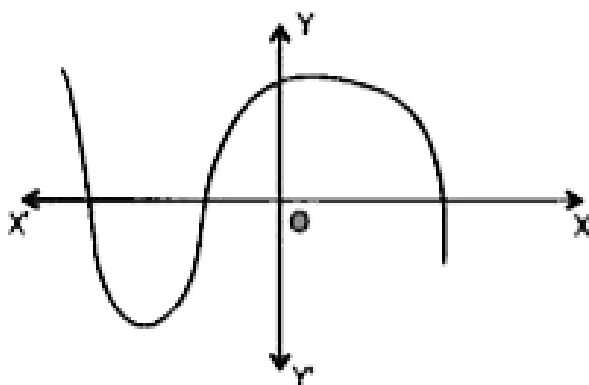
a.  $7 \times 11 \times 13$

b.  $3 \times 7 \times 11 \times 13$

c.  $7 \times 11 \times 23$

d.  $2^2 \times 3 \times 7$

2. Write the number of zeros of the polynomial, Whose graph is given below.

3. Create a linear equation which is parallel to the given line  $3x + 2y + 8 = 0$ .4. The **30<sup>th</sup>** term of the **A.P , 7, 4, .....**is

[     ]

A. 87

B. 77

C. -77

D. -87

5. If the length of a shadow of a pole is equal to the length of the pole then find the angle of elevation.

6. If  $P(E) = 0.05$ , then  $P(\bar{E}) =$  \_\_\_\_\_7. If  $\alpha$  and  $\beta$  are zeroes of the zeroes of the polynomial  $2x^2 - 13x + 6$ , then  $\alpha + \beta$  is equal to

[     ]

A. -3

B. 3

C.  $\frac{13}{2}$

D.  $-\frac{13}{2}$

8. If  $\tan \theta = \frac{4}{3}$ , then value of  $\sin \theta$  is

[     ]

A.  $\frac{3}{4}$

B.  $\frac{3}{5}$

C.  $\frac{4}{5}$

D.  $\frac{5}{3}$

9. Draw a rough sketch showing minor segment by shading.

[     ]

10. The shape of an ice cream cone forms a combination of \_\_\_\_\_ and \_\_\_\_\_

11. If  $\triangle ABC \sim \triangle EFD$  and  $\angle A = 40^\circ$ ,  $\angle F = 80^\circ$ , then find  $\angle C =$  \_\_\_\_\_

12. Write the standard form of a quadratic equation.

**II. Answer all the questions.****8 x 2 = 16m**

13. Define similar triangles.

14. Find the coordinates of the point which divides the join of **(-1,7)** and **(4, -3)** in the ratio **2: 3**.

15. "A person is flying a kite at an angle of elevation ' $\alpha$ ' and the length of thread from his hand to kite is ' $l$ '. Draw diagram for the situation.

16. Evaluate  **$2 \tan^2 45 + \cos^2 30 - \sin^2 60^\circ$** .

17. Find a quadratic polynomial whose sum and product of its zeroes are **-1** and  **$\frac{1}{4}$**  respectively.

18. Find the total surface area of a hemisphere of radius **7cm**.

19. Prove that the tangents drawn at the end of a diameter of a circle are parallel.

20. Find the value of **K** for the quadratic equation  **$9x^2 - kx + 4 = 0$** . So that this has two equal roots.

**III. Answer all the questions.****8 x 4 = 32m**

21. One card is drawn from a well-shuffled deck of 52 cards. Calculate the probability that the drawn card will be a king? Now create 4 such type of questions.

22. Write the formula for mode of a grouped data and explain each term of it.

23. A solid is in the shape of a cone standing on hemisphere with both their radii being equal to 1cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of  $\pi$ .

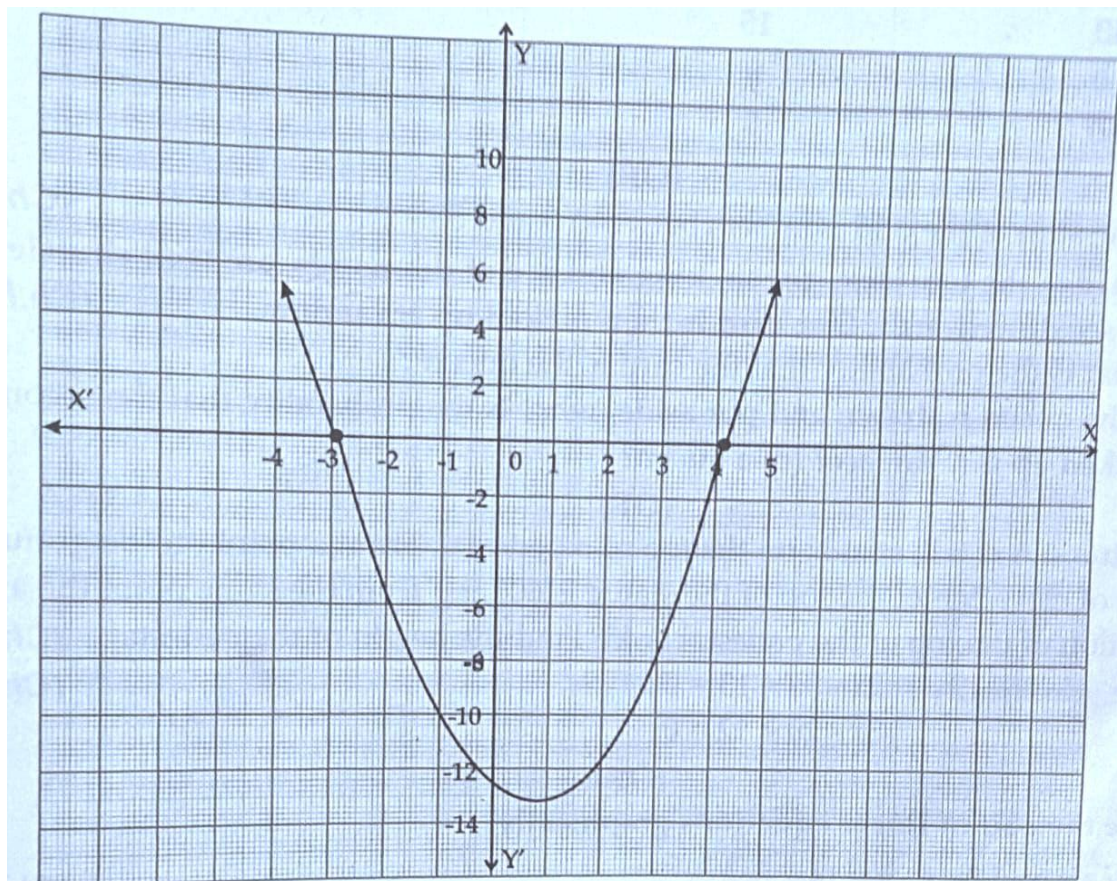
24. If  $\tan(A + B) = \sqrt{3}$ , then  $(a - B) = \frac{1}{\sqrt{3}}$ ,  $0^\circ < A + B < 90^\circ$ , then find A and B.

25. Determine the A.P whose 3<sup>rd</sup> term is 5 and 7<sup>th</sup> term is 9.

26. Find two numbers whose sum is 27 and product is 182.

27. A quadrilateral ABCD is drawn to circumscribe a circle prove that  **$AB + CD = AD + BC$** .

28. Answer the following questions based on the given graph.



- Name the shape of the graph.
- How many zeroes are there for the polynomial?
- The zeroes of the polynomial are \_\_\_\_\_
- Sum of zeroes of the polynomial \_\_\_\_\_

IV. Answer all the questions. There is an internal choice for each question.

5 x 8 = 40m

29. A. is  $\sqrt{3}$  irrational? Justify your answer.

(Or)

B. The diagonal of a quadrilateral **ABCD** intersect each other at the point O such that

$$\frac{AO}{BO} = \frac{CO}{DO}. \text{ Show that } \mathbf{ABCD} \text{ is a trapezium.}$$

30. A. Show that the quadrilateral formed by joining the points **(-4,2)**, **(4,4)**, **(2,12)** and **(-6,10)** taken in order is a square.

(Or)

B. A chord of a circle of radius 10cm subtends a right angle at the center. Find the area of corresponding.

i. minor segment.

ii. major segment (use  $\pi = 3.4$ )

31.A. If the median of the distribution given below is 28.5. Find the value of X and Y.

| Class interval | Frequency |
|----------------|-----------|
| 0-10           | 5         |
| 10-20          | X         |
| 20-30          | 20        |
| 30-40          | 15        |
| 40-50          | Y         |
| 50-60          | 5         |
| <b>Total</b>   | <b>60</b> |

(Or)

B. The first and the last terms of an AP are 17 and 350 respectively. If the common difference is 9, how many terms are there and what is their sum?

32. A. On card is drawn from a well shuffle deck of 52 cards. Find the probability of getting.

- i. a king red colour
- ii. a face card

- iii. a red face card
- iv. the jack of hearts

(Or)

B. A statue, **1.6m** tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is **60°** and from the same point the angle of elevation of the top of the pedestal is **45°**. Find the height of the pedestal.

33.A. Solve the following system of equation.  $x - 2y = 0$ ;  $3x + 4y = 20$ .

(Or)

B. Draw the graph of the equation  $x - y + 1 = 0$  and  $3x + 2y - 12 = 0$ . Determine the coordinates of the vertices of the triangle formed by the lines and the x-axis and shade the triangular region.